Lecture #4: explain MOE Architecture in Detail

| Gemma

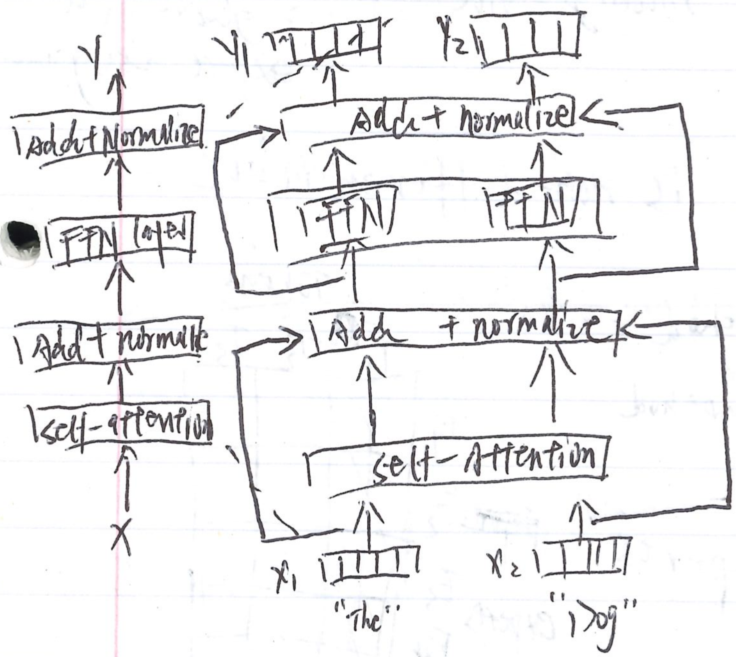
MoE: deepseek | Misteral | Grok | LLama4 | OMOEL | GPT4 ?

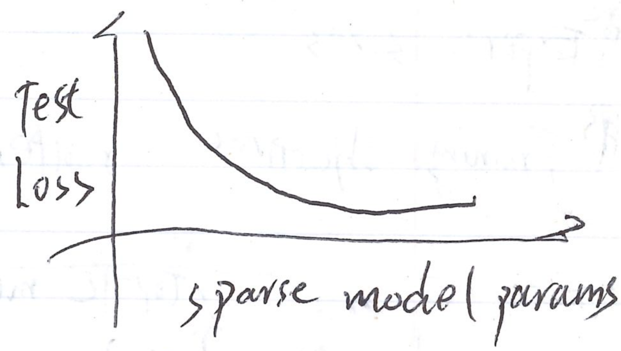
⇓ replace FFN with smaller FFN. and a selector layer.

{ increase #MoEs without affecting FLOP's
more FLOP's, but same FLOP | same perf

⇓

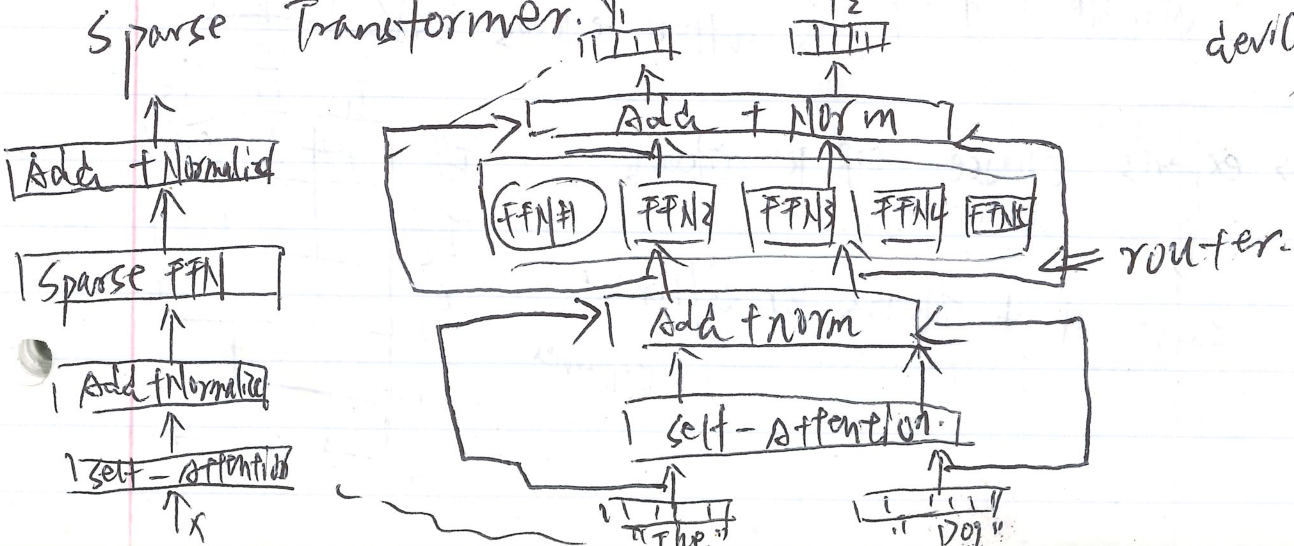
Dense Transformer.





⇑
{ Faster to Train
parallelize MoE to different devices

Sparse Transformer.



why haven't MoEs been more popular?

⇒ infrastructure is complex / advantages on multi-node

⇒ Training objectives are somewhat heuristic

MoE:

#1 routing → function (selector) = {
- ✓ token chooses experts: top-k ~~token~~ expert preferences
- Expert chooses tokens: top-k ~~expert~~ token preferences
- Global routing via opti: globally decide expert assignment.

#2 Expert sizes

#3 Training objectives: router is non-differentiable.

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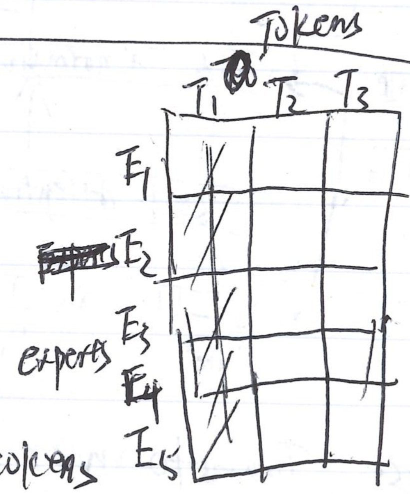
routing function: {
- Top-K method
- Hashing method

~~token~~

⇒ token choose top-k experts

⇓

variable # of Experts for different tokens



⇒ experts choose top-k tokens

⇓

same # of Tokens for different experts